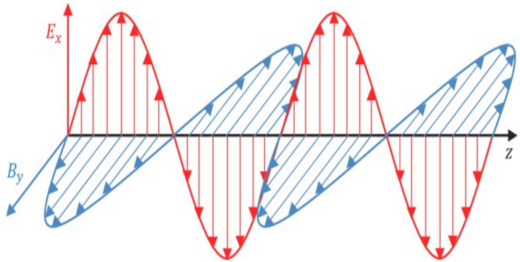

Integrated Sensing and Communication: Near/Far-Field Approaches in Multi-static Configurations

Saeid K. Dehkordi | Technische Universität Berlin | September 2023

Background and Preliminaries

- Beyond 5G and 6G wireless systems $\longrightarrow$ **ISAC** identified as key technology.
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  - Main goal $\longrightarrow$ design systems sharing the same frequency and hardware resources.

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- Typically large arrays are used in the mmWave-band to enable directional beamforming.
- We consider a HDA architecture at Tx/Rx, with multi-block processing.
- Large arrays and higher frequencies lead to an extended **Near-field** region.

$$\text{Fraunhofer distance:} \quad D_{ff} = \frac{2D^2}{\lambda}$$

- Before this distance, wave propagation adheres to the **spherical** model.
- In this model, the array response is a function of both angle and distance.

Array response model

$$\mathbf{a}(\phi_k, r_k) = \begin{pmatrix} \frac{r_k}{r_0} \exp(-j\frac{2\pi f_c}{c}(r_0 - r_k)) \\ \frac{r_k}{r_1} \exp(-j\frac{2\pi f_c}{c}(r_1 - r_k)) \\ \vdots \\ \frac{r_k}{r_{N_{\mathrm{tx}}-1}} \exp(-j\frac{2\pi f_c}{c}(r_{N_{\mathrm{tx}}-1} - r_k)) \end{pmatrix}$$

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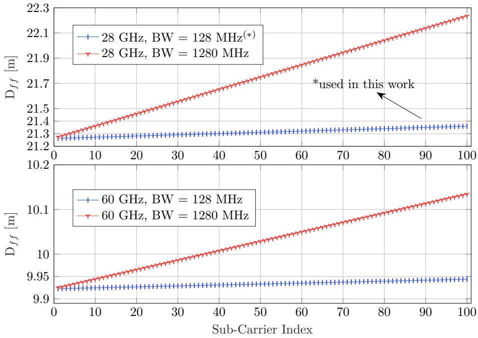
- In the far-field, i.e. $r \to D_{ff}(\approx \infty)$, then the well-known FF array response is obtained.

$$r_i - r_k \approx \frac{(id\cos\phi_k)^2}{2r_k} - id\sin\phi_k \tag{1}$$

$$\mathbf{a}(\phi_k) = [e^{-j\frac{N_{\mathrm{tx}}-1}{2}\pi\sin\phi_k}, \ldots, e^{j\frac{N_{\mathrm{tx}}-1}{2}\pi\sin\phi_k}]^T. \tag{2}$$

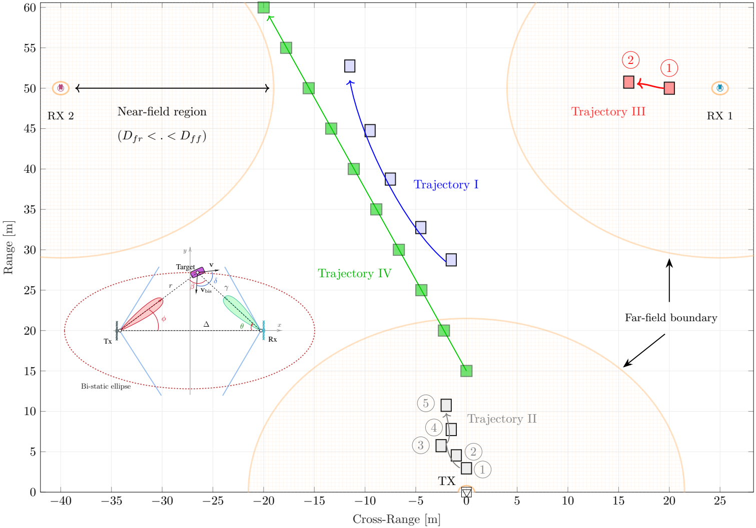
Near-Field Modeling: Multi-band systems

- In multi-band (multi-carrier) systems the Fraunhofer distance is a function of the frequency band.
- e.g. for OFDM, the Fraunhofer distance of the highest frequency sub-carrier, i.e. $D_{\text{ff}} = 2D^2/\min(\lambda_m)$ is the overall FF.



System Configuration

Trajectory 1 | Trajectory 2 | Trajectory 3 | Trajectory 4



Multi/Bi-static Sensing

- Bi-static configuration eliminates the need for full-duplex operation.
- Enables the possibility of mobile transmitters/receivers.
- Potentially improved target detection due to multiple observation angles.

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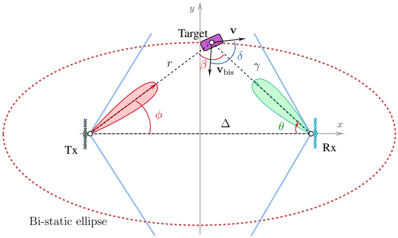
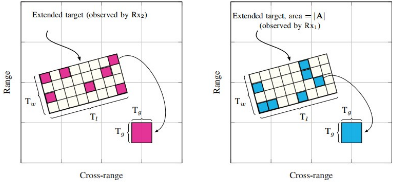


- Note: Multi-static can be defined as Tx/multiple Rx, multiple Tx/multiple Rx, · · · , i.e. any combination of bi-static pairs.

Target Modeling

- Motivated by **FEM** methods we use a grided Binomial Distribution to model the target.

$$\mathbb{P}(P = p) = \binom{|\mathcal{P}|}{p} q^p (1-q)^{|\mathcal{P}|-p}$$



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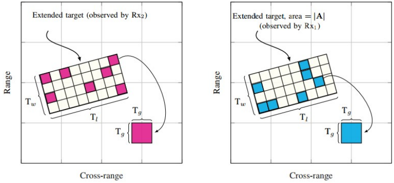

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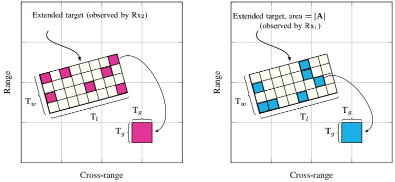

- Potentially improved target detection due to **multiple observation angles**.
- For a very large number of elements on the grid, i.e. $|\mathcal{P}| \to \infty$, each of which is independent active or non-active, the BND can also be very well approximated by a PPP, with intensity $\gamma = q|\mathcal{P}|$.

$$\mathbb{P}(P = p) = \frac{(\gamma|\mathbf{A}|)^p}{p!} e^{-\gamma|\mathbf{A}|}$$

Parameter Estimation

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  - Stage 1: FF beamforming and estimation based on FF bi-static assumptions.
  - Stage 2: If necessary, beamfocusing and estimation based on NF model.

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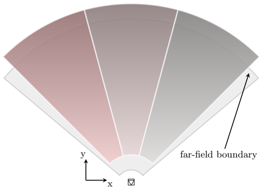

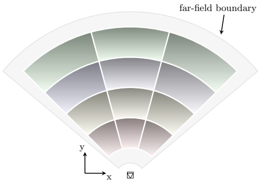

- The received signal at a given receiver array is given by:

$$\underline{\mathbf{y}}_b = \sum_{p=0}^{P-1} \mathring{\varepsilon}_p \mathbf{G}_b(\mathring{\nu}_p, \mathring{r}_p, \mathring{\gamma}_p, \mathring{\theta}_p, \mathring{\phi}_p)\underline{\mathbf{x}}_b + \mathbf{w}_b, \tag{3}$$

- where the space is of the ML problem is $\Gamma \triangleq \mathbb{C}^P \times \mathbb{R}^{5P}$!

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Stage 1:

- FF beamforming codewords at the Tx are selected from a codebook to illuminate a specific beam space segment.
- The simplified FF model of (3) is evaluated at each Rx for parameter estimation.

$$\underline{\mathbf{y}}_b = \sum_{p=0}^{P-1} \mathring{h}_p \breve{\mathbf{G}}_b(\mathring{\nu}_p, \mathring{\tau}_p, \mathring{\theta}_p)\underline{\mathbf{x}}_b + \mathbf{w}_b, \tag{4}$$

- where the space is $\Gamma_{\mathrm{FF}} \triangleq \mathbb{C}^P \times \mathbb{R}^{3P}$.

Parameter Estimation: continued $\cdots$

- If the target is in the NF, the model in (4) is **mismatched**.
  - However, the results are used to define a region of interest for illumination and estimation with the exact model.

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Stage 2:

- If target is determined to be in the NF of the Rx $\longrightarrow$ evaluate the model in (3) inside a RoI.
- Follows the grided ML framework with hypothesis testing with an adaptive threshold on radar image.

$$\ell(\nu, r, \gamma, \theta, \phi) = \frac{\left| \mathbf{f}^{\mathsf{H}} \mathbf{a} \mathbf{b}^{\mathsf{H}} \boldsymbol{\xi}_{(B)}(r, \gamma, \nu) \right|^2}{\mathbf{f}^{\mathsf{H}} \mathbf{a} \mathbf{b}^{\mathsf{H}} \bar{\mathbf{U}}_{(B)} \mathbf{b} \mathbf{a}^{\mathsf{H}} \mathbf{f}}. \tag{5}$$

- FF beamformer at the TX is sufficient since target is in FF of transmitter.

Parameter Estimation: continued ···

Stage 2:

- If the target is determined to be in the NF of the Tx $\longrightarrow$ A **beamfocusing** codeword needs to be selected.
- Parameter estimation at Rx follows the grided ML framework with hypothesis testing with an adaptive threshold on radar image with *reduced* FF model.

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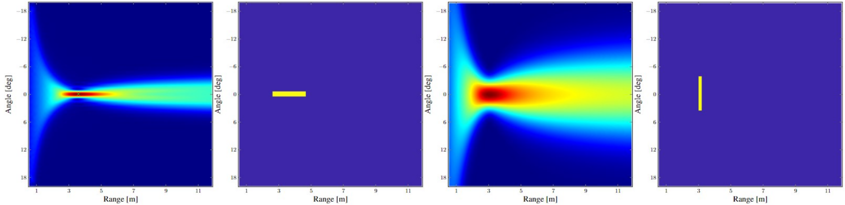
- Important to note that conventional beamfocusing schemes use the conjugated array response model $\rightarrow$ very sensitive to the exact parameters i.e. range and angle
- For user devices that are physically larger $\rightarrow$ antenna could be located anywhere on the object!
- The solution could be to use beamfocusers! that illuminate an extended region with a relatively constant gain.

Beamfocusing Design with Magnitude Least Squares



[1] P. W. Kassakian, “Convex approximation and optimization with applications in magnitude filter design and radiation pattern synthesis”, PhD dissertation, University of California, Berkeley, 2006

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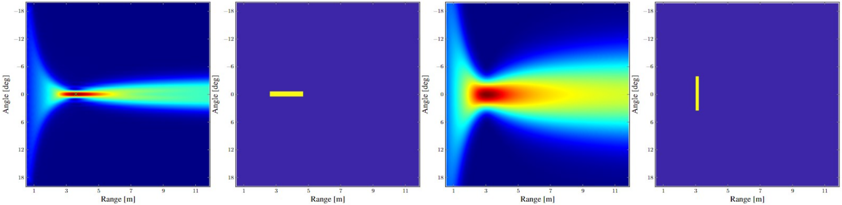

Magnitude Least Squares[1]

$$\min_{\mathbf{f}} \quad \| \, |\mathbf{A}^H \mathbf{f}| - \bar{\mathbf{b}} \|_2^2$$

$$\text{s.t.} \quad \mathbf{f}^H \mathbf{A} \mathbf{A}^H \mathbf{f} = 1$$

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Numerical Results

Far-field estimation performance (Trajectory I): Topology overview

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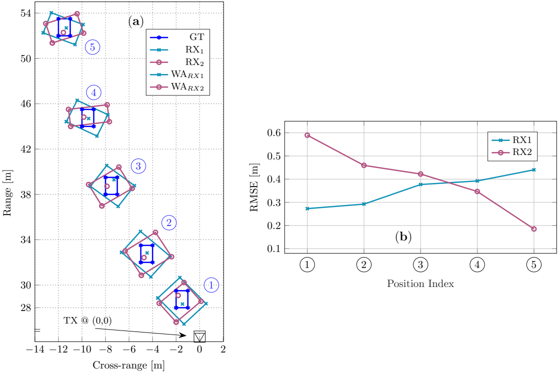
Topology overview



Multi-static spatial diversity gain (Trajectory IV)

Topology overview

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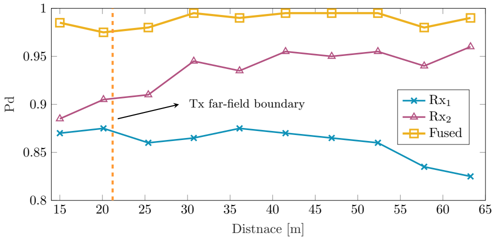

- Each target realization is performed according to the BND separately for each Rx.
- The *Fused* curve, is the average detection probability if at each step, either of the Rxs has detected the target (i.e. OR operation).
- This fusion can be performed at a central node.

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Estimation of targets in receiver NF (Trajectory III)

Topology overview

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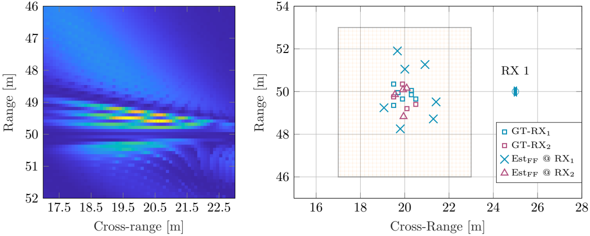
▶▶ Topology overview



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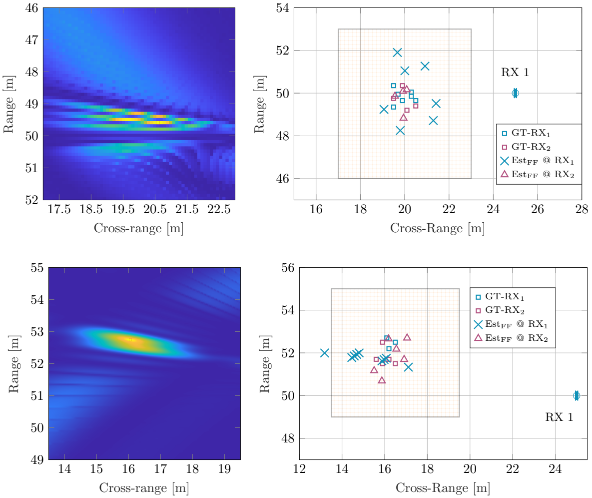

Spectral efficiency enhancement with beamfocusing (Trajectory II)

Topology overview

Numerical Results

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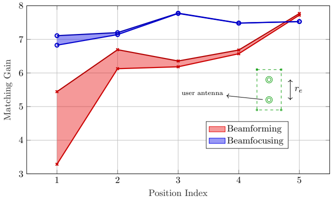
Topology overview

$$G_{\mathrm{M}} = \left|\mathbf{a}^{H}(r_0, \phi_0)\mathbf{f}_{\mathrm{X}}(\hat{r}, \hat{\phi})\right|, \quad \mathbf{f}_{\mathrm{X}} \in \{\mathbf{f}_{\mathrm{NF}}, \mathbf{f}_{\mathrm{FF}}\} \tag{6}$$



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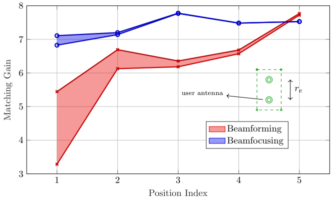

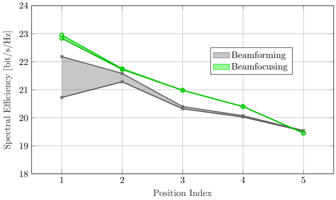

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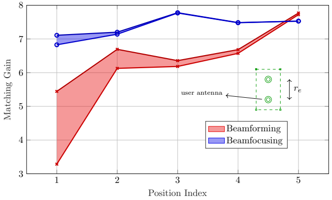
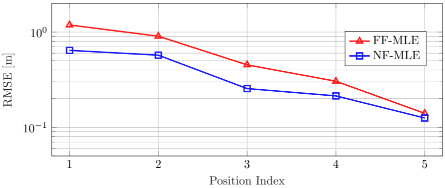
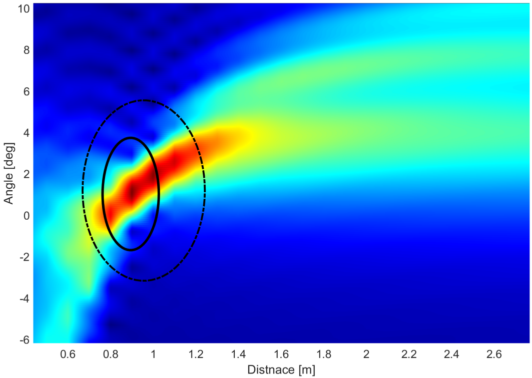




Thank you for your attention!

Questions?

Back-up slides



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